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The Multiple-Access Channel with Partial State Information at the Encoders

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Abstract

This work studies the multiple access channel controlled by random parameters, with full side information at the decoder, and partial, rate limited, side information (SI) at the encoders. A single letter characterization of the capacity region is derived, for the special case where the side information is degraded. Here *degraded* side information refers to the case where the SI available at one of the encoders is a subset of the SI available at the other decoder. Inner and outer bounds are derived on the capacity region of that channel, for the general case where there are no restrictions on the structure of the SI at the two encoders. The techniques employed for coding the rate-limited SI, and the achievable regions so obtained, are closely related to the problems of hierarchical source coding, and multiple descriptions.

Index terms — Hierarchical coding, multiple access channel, multiple description, side information, state controlled channels, successive refinement.

1 Introduction

Channels that are controlled by random states are widely studied from both information theoretic and communications aspects. Beside their theoretical importance, channels with random parameters model faithfully the situation in practical systems, where the random